Assignment Brief
Working with IT Software Spreadsheet and Database
Academic Year 2026

Unit Information:	
Qualification:	**Foundation Diploma in Business / CS & IT (SCQF Level 6 – Equivalent to A Level)**
Unit Code & Title:	**F1FJ 12** **Working with IT Software Spreadsheet and Database**
Assignment Title:	**Assignment 1** LO3 - Task A: Report (Individual) LO4 - Task B: Report (Individual)
Results Reporting:	Total Marks: 70 Assignment 1: Task A: 40% Assignment 1: Task B: 30%

Date of Issue: 29-01-2026	**Due Date:** 15-02-2026

To be Filled by the student

Student USN:

College Name & Site: JAIN College

Tutor:

Date Due:

Date of Submission:

*All work must be submitted on or before the due date. If an extension of time to submit work is required, a Mitigating Circumstance Form must be submitted.

Has an extension been approved? ☐ **Yes** ☐ **No**

If yes, please provide the new submission date ……/….. /……., and affix appropriate evidence.

First Marker:	**Second Marker:**
Agreed Mark:	**Refer: Yes / No**

General Guidelines for Assessment

1. Attach a title page to your assignment. Use the previous page as your cover sheet and ensure that you fill in the details correctly.
2. All assignment work should be prepared using word processing software and should be organized and clearly structured

Word Processing Guidance

1. Use a font type that makes it easy for your examiner to read, for example, Arial or Times New Roman.
2. Use 1.5-line line spacing. Left-justify all paragraphs.
3. Ensure that all headings are consistent in terms of size and font style.
4. **Use the footer function to insert your Student USN, Name, Unit Code, Unit Title, and Page Number on each page**.
5. Use the spell check and grammar check functions to check your assignment.

Assignment Submission

1. Ensure that you give yourself enough time to complete the assignment by the due date. You must take responsibility for managing your own time effectively.
2. Submit an electronic version uploaded to the student learning management system on or before the stated deadline.
3. Submissions received up to 72 hours after the deadline will be penalized in line with JAIN College's Late Submission Policy. Thereafter, submissions will not be accepted.
4. If you are unable to hand in your assignment on time and have valid reasons, such as illness, you may apply (in writing) for an extension under the terms of the JAIN COLLEGE Mitigating Circumstances Policy.
5. Non-submission of work without a valid reason will lead to an automatic REFERRAL. You will then be asked to complete an alternative assignment. Your work will be capped at a PASS.

Declaration of Authenticity

6. Exercise meticulous care when incorporating the work or ideas of others into your assignment. Ensure that all sources are accurately acknowledged using the Harvard referencing system, both within the text and in the bibliography. Failure to provide proper attribution may result in an investigation and potential penalties for plagiarism, in accordance with the JAIN College Malpractice Policy

Statement of Originality and Student Declaration

I hereby declare that I fully understand the concept of plagiarism, which includes the use of another individual's work and its presentation as my own without proper and accurate attribution of sources. I further acknowledge my awareness of what constitutes copying or the use of another's work without appropriate academic acknowledgement.

1. I know that plagiarism is a punishable offence because it constitutes theft, and the consequences of plagiarism as outlined in the Malpractice Policy.
2. I understand JAIN College's plagiarism, ghost writing and copying policy.
3. I declare that all work presented by me for every aspect of this assignment is my own, and where I have made use of another's work, I have rightly attributed the source.
4. I acknowledge that the attachment of this document, signed or not, constitutes my agreement to the above.
5. I understand that my assignment will not be considered as submitted if this declaration is not included.

Student's Signature: …… **Date: ……**

Assignment 1: Report on Working with IT Software Spreadsheet and Database

The assessment measures the attainment of learning outcomes through one summative task.

Task A – Summative Assessment (LO3)

You are required to submit an individual report for the given assessment. Your report should cover the following points:

Use Advanced Spreadsheet and Database Functions to solve tasks

1.1 Use advanced spreadsheet features such as filtering, goal seek, charting, trend analysis, lookup functions, and database functions to analyze data. **(3 + 3 + 4 = 10 Marks)**

Sales_Expenses_Dataset

Month	Product	Region	Sales	Expenses
Jan	Alpha	North	120381	27954
Jan	Beta	South	91346	45202
Jan	Gamma	East	102270	65063
Feb	Alpha	South	93119	65176
Feb	Beta	East	94992	27430
Feb	Gamma	East	111759	50661
Mar	Alpha	East	100309	63244
Mar	Beta	North	117702	59276
Mar	Gamma	South	92088	30908
Apr	Alpha	North	107582	58951
Apr	Beta	East	66726	35949
Apr	Gamma	North	91451	44425
May	Alpha	South	132176	32773
May	Beta	East	63943	20157
May	Gamma	South	130521	55242
Jun	Alpha	North	98244	59716
Jun	Beta	West	112039	46328
Jun	Gamma	East	102008	62922
Jul	Alpha	West	93706	54516
Jul	Beta	West	142335	25756
Jul	Gamma	South	105721	61854
Aug	Alpha	West	121352	65223
Aug	Beta	South	121201	41054
Aug	Gamma	East	145179	31784
Sep	Alpha	East	79130	23412
Sep	Beta	North	123666	24990
Sep	Gamma	East	63837	51356
Oct	Alpha	East	91223	64400

Oct	Beta	East	133792	34399
Oct	Gamma	West	139985	39846
Nov	Alpha	East	142135	74649
Nov	Beta	North	64497	66877
Nov	Gamma	West	67123	61090
Dec	Alpha	North	135742	49183
Dec	Beta	West	121123	29697
Dec	Gamma	West	112899	76234

Dataset: Sales_Expenses_Dataset (LO_3.1)

1.1.1 Filtering

Filter the dataset to show only Product Alpha in the North region for any 3 months.

1.1.2 Chart Creation

Create a column chart comparing Sales vs Expenses for all products in January.

1.1.3 Pivot Table

Create a Pivot Table summarizing:

- Total Sales
- Total Expenses

by:

- Product
- Region

Add a Slicer for Product.

1.2 Apply at least three advanced spreadsheet tools, such as templates, data validation, macros, worksheet protection, and pivot tables.

(5 + 5 + 5 = 15 Marks)

Dataset: Employee Training

EmployeeID	EmployeeName	Department	TrainingProgram	Hours	Cost
E101	John Smith	HR	Communication Skills	12	15000
E102	Emma Johnson	Finance	Excel Advanced	16	18000
E103	Liam Brown	IT	Cloud Fundamentals	20	25000
E104	Olivia Davis	Marketing	Digital Marketing	14	20000
E105	Noah Wilson	HR	Team Building	10	12000
E106	Ava Martinez	Finance	Financial Modelling	18	22000
E107	Ethan Miller	IT	Cybersecurity Basics	22	27000
E108	Sophia Taylor	Marketing	Branding Strategy	16	21000
E109	Lucas Anderson	HR	Conflict Management	12	14000
E110	Mia Thompson	IT	AI Essentials	24	30000

1.2.1 Data Validation and Dynamic Summary

a. Create a dropdown list of departments using **ValidationLists → DepartmentList**.

b. Based on the selected department, build a **dynamic summary table** showing:

1. Total Training Hours
2. Total Training Cost
3. Number of Employees Trained

1.2.2 Macro Recording

Record a macro named **FormatTrainingReport** that performs the following:

a. Boldens all headers

b. Applies currency format to the Cost column

- Autofits all columns
- Highlights the header row with a light fill color

1.2.3 VLOOKUP

Using VLOOKUP, retrieve the Training Program for all employees after copying Employee Names into a separate table.

1.3 Create and use database (MS Access) objects, including tables and queries (sorting, expression-based, update/append/delete) using multiple tables.

(10 + 5 = 15 Marks)

Dataset

Table 3: Borrowing

Field Name	Type
BorrowID	AutoNumber (PK)
MemberID	Number (FK)
BookID	Number (FK)
BorrowDate	Date
ReturnDate	Date
DaysBorrowed	Number

BorrowID	MemberID	BookID	BorrowDate	ReturnDate	DaysBorrowed
201	101	1	1-Jan-25	9-Jan-25	8
202	102	4	5-Jan-25	12-Jan-25	7
203	103	2	10-Jan-25	20-Jan-25	10
204	105	3	15-Jan-25	22-Jan-25	7
205	104	5	18-Jan-25	27-Jan-25	9

1.3.1 Create a Data Entry Form with Form Controls

Create a form named **BorrowForm** based on the Borrowing table.
Form Must Include:

a) **Text Boxes**
- BorrowID
- BorrowDate
- ReturnDate
- DaysBorrowed

b) **List Box**
- Displays list of all Book Titles from the Books table
- User selects a title to assign a BookID

c) **Labels**
- For each field (properly aligned and formatted)

Add Command Buttons:

Include the following buttons (wizard allowed):
- Add New Record
- Delete This Record
- Go to First Record
- Go to Last Record

1.3.2 Write Queries Using Sorting and Expressions

Sorting Query

Create a query SortedBooks that:

- Lists all books
- Sorted by Genre (Ascending) and then Price (Descending)

Task B – Summative Assessment (LO4)

LO4: Illustrate and evaluate solutions using appropriate software

Task 1- Select the most suitable application (spreadsheet or database) for a specified task and justify the choice.

- **Excel Task B (2.0)** **(15 Marks)**
  Dataset: StoreData

Month	Electronics	HomeGoods	Clothing	Elec_Expenses	Home_Expenses	Cloth_Expenses
January	120000	85000	60000	70000	40000	30000
February	135000	90000	65000	75000	42000	31000
March	150000	95000	70000	82000	45000	33000
April	160000	100000	72000	88000	47000	34000
May	170000	99000	75000	92000	46000	35000
June	155000	97000	77000	83000	45500	36000

This task involves analysing and presenting store performance data using Microsoft Excel to meet the needs of different business audiences. Learners will work with the *StoreData* datasets, which contain six months of sales and expense information for key product categories. Through the creation of manager-focused dashboards, accountant-level detailed tables, and a written reflection, learners will demonstrate their ability to select suitable Excel features, apply complex calculations, present information clearly, and tailor outputs to audience requirements. The activities reflect real business reporting scenarios where visual summaries, accurate analysis, and professional presentation are essential for decision-making across departments.

2.1 Explain why Microsoft Excel is the most suitable application for producing the Managers' Summary Report and the Accountant's Detailed Report.
(Consider charts, calculations, audience needs, and visual presentation.) **(3 Marks)**

2.2 Execute the tasks below in Excel and provide screenshots or exported outputs as evidence.
(3+3+3 =9 Marks)

2.2.1 Managers' Summary Report

Total Sales per Product (Chart)

Create a **column chart** comparing total 6-month sales for:

- Electronics
- HomeGoods
- Clothing

2.2.2 Generate Two (2)-Month Forecast

For Electronics:

- Create a line chart of monthly sales
- Add a Linear Trendline

- Display the following:
    - Equation
    - R^2 value
    - Forecast forward 2 periods

2.2.3 Create Accountant's Detailed Report

Create a detailed table including:

- Month, Product, Sales, Expenses, Profit (Profit = Sales – Expense (formula visible))
- SUMIF to calculate total sales per product
- IF to flag months with Profit < 20,000

Audience: Accounting team → need detail, accuracy, formulas visible.

2.3 Evaluate the strengths and limitations of the Excel reports that you have created. Elaborate on presentation quality, accuracy, usefulness to the target audience, and possible improvements for future efficiency. **(3 Marks)**

Database Task B (3.0) **(15 Marks)**

Hospital Patient Management_LO_4

Dataset

Table 1 — Patients

Field	**Type**
PatientID	AutoNumber (PK)
FirstName	Short Text
LastName	Short Text
Age	Number
Department	Short Text
AdmissionDate	Date

PatientID	FirstName	LastName	Age	Department	AdmissionDate
1	John	Miller	45	Cardiology	02-Jan-25
2	Emma	Collins	32	Neurology	05-Jan-25
3	Lucas	Brown	58	Orthopedics	07-Jan-25
4	Sophia	Green	40	Cardiology	10-Jan-25
5	Ethan	White	29	Pediatrics	12-Jan-25

Table 2 — TreatmentRecords

Field	**Type**
RecordID	AutoNumber (PK)
PatientID	Number (FK)
TreatmentType	Short Text
Cost	Currency
DaysAdmitted	Number
DischargeDate	Date

RecordID	PatientID	TreatmentType	Cost	DaysAdmitted	DischargeDate
101	1	Angiography	35000	4	06-Jan-25
102	2	Brain MRI	18000	2	07-Jan-25
103	3	Joint Replacement	52000	10	17-Jan-25
104	4	Stress Test	8000	1	11-Jan-25
105	5	Child Wellness	5000	1	13-Jan-25

This task focuses on designing and evaluating a small-scale Hospital Patient Management System using Microsoft Access. Learners will work with linked tables containing patient details and their treatment records to create a functional database solution. Using forms, queries, and reports, learners will demonstrate their ability to select the most appropriate application, produce accurate and well-structured outputs, and evaluate the effectiveness of their work. The activities simulate real hospital workflows such as entering treatment information, searching across multiple tables, and generating summary reports for administrative decision-making.

3.1 Explain why MS Access is the most appropriate application for creating the *Patient Treatment Entry Form*, compared to using a spreadsheet. (Consider features like data entry, relationships, and navigation controls.) **(3 Marks)**

3.2 Execute the tasks below in MS Access and provide screenshots or exported objects as evidence. **(5 + 5 = 10 Marks)**

3.2.1 Create a Patient Treatment Entry Form

Using the TreatmentRecords table from the Hospital database, create a form named TreatmentEntryForm with the following:

a) Required Form Objects

Text Boxes for:

- RecordID
- TreatmentType
- Cost
- DaysAdmitted
- DischargeDate

List Box

- Shows Patient Names from the Patients table
- When a patient is selected, their PatientID field is filled automatically

Labels for all fields

- Professionally aligned
- Clear headings such as “Patient Name”, “Days Admitted”, etc.

b) Required Command Buttons (Using Button Wizard)

Add functional buttons to the form:

- Add New Record

3.2.2 Multi-Table Search Query

Create a query named PatientTreatmentSummary using:

- Patients
- TreatmentRecords

The query must: Join all related tables

- Patients.PatientID → TreatmentRecords.PatientID

Required Fields

- Patient Name
- Department
- TreatmentType
- Cost
- DaysBorrowed
- AdmissionDate
- DischargeDate

3.3 Evaluate the strengths and limitations of the form, queries, and report you created. Determine what aspects worked well, and suggest what improvements could be made for greater accuracy, efficiency, or usability? **(2 Marks)**

Note:

- All learners are required to mandatorily participate in both Task A and B of Assignment 1 (Summative Task).
- The assignment will be assessed based on the single report submitted for both Task A and B of Assignment 1 (Summative Task).
- Your reports should cover all the learning outcomes and the assessment criteria.
- To understand the marking scheme, please refer to the marking criteria.

Marking Criteria

Learning Outcome	Task	Assessment Marking Criteria	Marks
LO3	1.1	Use advanced spreadsheet features such as filtering, goal seek, charting, trend analysis, lookup functions, and database functions to analyze data	10
	1.2	Apply at least three advanced spreadsheet tools such as templates, data validation, macros, worksheet protection, and pivot tables.	15
	1.3	Create and use database objects including tables and queries (sorting, expression-based, update/append/delete) using multiple tables. .	15
LO4	2.1	Explain why Microsoft Excel is the most suitable application for producing the Managers' Summary Report and the Accountant's Detailed Report.	03
	2.2	Execute the tasks below in Excel and provide screenshots or exported outputs as evidence.	09
	2.3	Evaluate the strengths and limitations of the Excel reports you created.	03
	3.1	Explain why MS Access is the most appropriate application for creating the Patient Treatment Entry Form, compared to using a spreadsheet.	03
	3.2	Execute the tasks below in MS Access and provide screenshots or exported objects as evidence.	10
	3.3	Evaluate the strengths and limitations of the form, queries, and report you created.	02
Total			70

The minimum marks required to pass this assignment is 40%